Title:
Enhancing ERC-4907, ERC-1155, and ERC-721 with AI for Real-World Asset Tokenization and NFT-Based Rentals

Summary :
This research explores real-world asset tokenization using ERC-4907, ERC-1155, and ERC-721 standards, focusing on both asset ownership and rental models. While ERC-721 and ERC-1155 facilitate asset tokenization, ERC-4907 introduces NFT-based rental functionalities. The study aims to enhance these standards by integrating granular usage controls, automated revenue-sharing mechanisms, and AI-driven fraud detection to optimize transactions from buying to renting tokenized assets, tracking and facilitating the process through Agentic AI. Additionally, this research will propose improvements to the Ethereum Foundation for enhancing these standards, ensuring seamless real estate tokenization, improved rental management, and broader adoption of decentralized leasing models.

Rationale - Review of Relevant Research / Limitations/Gaps:
Real-world asset tokenization has gained traction through blockchain standards like ERC-721 and ERC-1155. However, these standards primarily focus on ownership and lack robust mechanisms for rentals. ERC-4907 introduced time-limited NFT rentals, but key gaps remain:

- Granular Usage Control: Existing models do not provide detailed conditions for asset utilization beyond ownership.
- Automated Revenue Sharing: No standardized mechanism for distributing rental income among stakeholders.
- Security and Fraud Detection: Risks such as unauthorized duplication and fraudulent rentals remain unaddressed.
- Agentic AI: AI-driven optimizations in rental pricing, fraud prevention, and smart contract security remain underexplored. I aim to propose improvements using Agentic AI to autonomously tokenize real estate assets, execute smart contracts, and manage fractional ownership, while also enhancing compliance, dispute resolution, and secondary market liquidity. As this is a broad scope, I will refine my focus on the use of Agenti AI as the research progresses.

This study addresses these gaps by integrating AI-powered automation and enhanced smart contract functionalities, ensuring a seamless transition from buying to renting real-world tokenized assets. Additionally, this research will contribute recommendations to the Ethereum Foundation to refine ERC-4907, ERC-1155, and ERC-721 for rental-based applications.

Significance of the Study:
This research will:

1. Enhance ERC-4907, ERC-1155, and ERC-721 by proposing improvements to their existing frameworks to enable seamless real-world asset tokenization, including rental functionalities. However, during the research, there might be a possibility that the existing proposals become a bottleneck to the proposed changes, in which case we may suggest a new proposal. This is yet to be ascertained, but our foremost goal will be to suggest changes to the existing standards.
2. Develop AI-driven fraud detection and security mechanisms for NFT-based asset rentals, leveraging Agentic AI to autonomously monitor transactions, detect anomalies, prevent fraudulent activities, and enhance overall efficiency in the process.
3. Improve rental monetization models with automated revenue-sharing mechanisms.
4. Promote greater adoption of real-world asset rentals via tokenized ownership and leasing models.
5. Provide recommendations to the Ethereum Foundation for enhancing NFT standards in real estate tokenization and rentals.

Research Questions:

1. How can ERC-4907, ERC-1155, and ERC-721 be extended to support real-world asset tokenization and rental applications?
2. What AI models can be integrated to enhance fraud detection and security in NFT-based rentals?
3. How can smart contracts automate fair and transparent revenue-sharing mechanisms for property rentals?
4. What are the economic and security implications of integrating AI-driven rental agreements?
5. How can improvements to ERC-4907, ERC-1155, and ERC-721 be proposed to the Ethereum Foundation to enhance NFT rental functionalities?

Intended Methodology and Project Feasibility:

1. Phase 1 – Data Collection (Year 1)
    - Collect transaction logs from NFT-based real estate and asset tokenization platforms.
    - Conduct surveys and interviews with real estate and rental stakeholders.
    - Identify fraud patterns and inefficiencies in current tokenized rental models.
2. Phase 2 – Model Development (Year 2)
    - Develop smart contract extensions for ERC-4907, ERC-1155, and ERC-721 tailored to real estate rentals.
    - Implement automated revenue-sharing logic in Solidity.
    - Train AI models for fraud detection and predictive pricing, process facilitation.

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    - Propose enhancements to ERC-4907, ERC-1155, and ERC-721 for broader Ethereum ecosystem improvements in real-world asset rentals.
3. Phase 3 – Implementation & Validation (Year 3)
    - Deploy test contracts on Ethereum testnets.
    - Validate AI models in live simulations.
    - Conduct pilot tests with NFT real estate and rental platforms.
    - Submit proposed ERC-4907, ERC-1155, and ERC-721 improvements to the Ethereum Foundation, emphasizing real-world asset rentals.

Brief Timeline (Over 3 Years):

Year 1: Foundation & Preliminary Research - Establish the theoretical and technical foundation for asset tokenisation and rental models.

Literature Review & Standards Analysis

- Study ERC-4907, ERC-1155, and ERC-721 standards in-depth.
- Review existing tokenisation projects, their challenges, and limitations.
- Analyse real-world asset tokenisation use cases (e.g., real estate, luxury goods).

Technical Feasibility & Smart Contract Development

- Evaluate the existing functionalities and identify gaps in current ERC standards.
- Develop prototype smart contracts for asset tokenisation and rental mechanisms.
- Explore granular usage controls and how they can be implemented.

Stakeholder & Regulatory Analysis

- Study regulatory implications of asset tokenisation and rental models.
- Identify legal frameworks for tokenised real estate and digital leasing.
- Engage Ethereum Foundation, and DeFi developers.

Year 2: Implementation & AI-Driven Optimisation - Develop and test AI-integrated enhancements for improved security and efficiency.

Building Automated Revenue-Sharing & AI Fraud Detection

- Implement automated revenue-sharing mechanisms in smart contracts.
- Integrate AI-driven fraud detection to prevent misuse of rental assets.
- Simulate transaction flows from tokenised asset purchase to rental execution.

Agentic AI for Process Tracking & Optimisation

- Design an Agentic AI system to track and optimise tokenised rental processes.
- Implement predictive analytics to assess tokenised asset demand and usage.
- Test AI-driven decision-making in lease management and asset tracking.

Deployment & Real-World Experimentation

- Deploy enhanced ERC prototypes on Ethereum.
- Conduct simulated asset tokenization and rental transactions.
- Gather data on security, efficiency, and user experience.

Year 3: Evaluation, Refinement & Thesis Submission - Finalise the research, validate the results, and submit findings.

Performance Evaluation & Refinement

- Conduct stress testing for scalability and security.
- Refine smart contracts based on user feedback and real-world testing.

Proposal to Ethereum Foundation

- Draft an Ethereum Improvement Proposal (EIP) with suggested ERC enhancements.
- Present findings at blockchain conferences and Ethereum developer forums.

Thesis submission

- Draft the research thesis, focusing on findings, ERC improvements, and AI integration.
- Peer reviews, revisions, and final thesis submission.

This research aligns with the growing demand for secure, automated, and Agentic AI-enhanced real-world asset tokenization and rental ecosystems, ensuring feasibility and long-term impact.

Research Output and Publication

1- Technical Whitepaper on ERC Standard Enhancements – A detailed whitepaper outlining the proposed improvements to ERC-4907, ERC-1155, and ERC-721, including technical justifications and smart contract design considerations.
2- Simulation and experimental results – Empirical validation of the proposed AI-enhanced NFT rental framework using real-world datasets, highlighting its efficiency in fraud detection, revenue sharing, and automated contract execution.
3- Ethereum Improvement Proposal (EIP) Draft – A formal draft proposal for the Ethereum Foundation to consider integrating suggested modifications into ERC standards.
4- Implementation Guide for Developers – A step-by-step guide for integrating Agentic AI into NFT rental systems, allowing developers to adopt and experiment with the improvements in real-world applications.
5- Presentation at Blockchain & AI Conferences – Submission of research findings to major blockchain and AI conferences (e.g., Devcon, NeurIPS, IEEE Blockchain) to gain industry feedback and adoption.

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